

Reports in Experimental physics 1-3

The report from an experimental investigation or a project in Experimental physics 1-3 typically covers 5-10 pages of text and additionally contains figures and appendices. The report is handed in as a pdf and can be prepared with a text editor of your own choice, for example word or LaTeX.

The report must be based on the **collected data** and the **logbook** from the investigation/exercise.

Content of a report

The report **must** include the following elements and sections:

- **Title**
- **Authors**
- **Introduction**
- **Experimental setup**
- **Results**
- **Discussion**
- **Conclusion**
- **References**
- **Appendices**

The mentioned sections should more specifically include:

- **Introduction**
 - Describe the purpose and ideas of the project/exercise.
 - Optionally describe the subject of the project/exercise in a broader context.
 - Explain briefly the central physical ideas for the project/exercise. Possibly support the brief explanation with figures.
 - Summarize the experimental methods applied in the investigation.
- **Experimental setup**
 - Describe the applied setup systematically.
 - Explain how measurements are performed.
 - Support the written text with figures/illustrations/photos, for example:
 - figure with a sketch of the experimental setup,
 - photo of the setup taken during the investigation
 - figures that show direct data (could e.g. be traces obtained with picoscope).
 - The descriptions must be so that a person who does not know the experimental system can repeat the experiments and potentially identify possible flaws.
 - Subsections can often be used to ensure a logical structure of the description.
- **Results**
 - Present measurements and data analyzes from the experimental investigation.

- Explain how the data has been evaluated, i.e., what calibrations, calculations, and analyzes (for example fits) have been done. Use figures and tables to clarify the data presentation.
- Always state uncertainties. For example, the results from a fit is always given with uncertainties and the fitted curve is often plotted together with the data.
- Apply relevant statistical methods.
- Reflect critically on the applied methods, i.e., consider for instance the quality of a fit.
- Consider sources of uncertainty and inaccuracy.
- **Discussion**
 - Discuss relevant aspects of the investigation. Consider for example
 - the agreement between measured data and theory,
 - reasons for deviations between experiment and theory,
 - possible improvements of the experiment
 - problems with assumption made in the experiment or theory
 - Consider using subsections to clarify the discussion.
- **Conclusion**
 - Give a short summary of the idea of the investigation and all relevant conclusions that have been obtained.
- **References**
 - Collect and number all references in order of appearance in the text.
 - Use a consistent format for the references, for example.
 - Authors, *Title*, Journal name, **vol**, pages (year).
 - Avoid as far as possible to make direct reference to wikipedia pages and unauthorized webpages, but try instead to find original literature and authorized databases.
- **Appendices**
 - Collect other relevant materials that belongs to the obtained data and performed analyzes, for example additional documentation of these.

Remark:

The written logbook is an essential part of the performed work and it must always be available for further documentation for example in cases of doubt about the pass/fail status of the report.